

UNIT II DISASTER RISK REDUCTION (DRR)

Sendai Framework for Disaster Risk Reduction, Disaster cycle - Phases, Culture of safety, prevention, mitigation and preparedness community Based DRR, Structural-non-structural measures, Roles and responsibilities of- community, Panchayati Raj Institutions / Urban Local Bodies (PRIs/ULBs), States, Centre, and other stakeholders- Early Warning System – Advisories from Appropriate Agencies. - Relevance of indigenous Knowledge, appropriate technology and Local resources.

Sendai Framework for Disaster Risk Reduction

The Sendai Framework for Disaster Risk Reduction (SFDRR) is a global agreement aimed at reducing disaster risks and increasing resilience. Adopted in March 2015 during the Third World Conference on Disaster Risk Reduction held in Sendai, Japan, it builds on previous frameworks and extends the global conversation on disaster risk management.

1. Purpose and Objectives

The Sendai Framework seeks to:

- **Reduce Disaster Risk:** Minimize the risk of disasters and mitigate their impacts on communities.
- **Strengthen Resilience:** Enhance the ability of societies to recover from and adapt to disasters.
- **Promote a Culture of Safety:** Foster awareness and education about disaster risk reduction (DRR).

2. Structure and Components

The Framework is structured around four priority areas, seven global targets, and four major priorities for action:

Priority Areas for Action

1. Understanding Disaster Risk:

- Encourage the development of risk knowledge by investing in data collection and analysis.
- Promote research and risk assessments to understand vulnerabilities and exposure to hazards.

2. Strengthening Disaster Risk Governance:

- Enhance institutional and legal frameworks for DRR.
- Ensure stakeholder engagement, including local communities and the private sector.
- Promote transparent and accountable decision-making.

3. Investing in Disaster Risk Reduction:

- Prioritize DRR in policies and planning.
- Encourage financial investments in risk reduction measures.
- Integrate DRR into development planning and strategies.

4. Enhancing Disaster Preparedness for Effective Response:

- Improve early warning systems and emergency response mechanisms.
- Foster community-based preparedness and response planning.
- Develop capacity-building initiatives for effective disaster management.

Global Targets

The SFDRR (Sendai Framework for Disaster Risk Reduction) outlines seven global targets to be achieved by 2030:

1. **Reduce Deaths:** Substantially reduce the number of people killed globally by disasters.
2. **Reduce Affected People:** Significantly reduce the number of people affected by disasters.
3. **Reduce Economic Losses:** Reduce economic losses relative to global GDP.
4. **Reduce Damage to Critical Infrastructure:** Reduce damage to critical infrastructure and disruption of services.
5. **Increase the Number of Countries with DRR Strategies:** Ensure more countries have national and local DRR strategies.
6. **Increase the Number of Countries with DRR Plans:** Increase the number of countries with DRR plans, policies, and institutional frameworks.
7. **Integrate DRR into National Policies:** Incorporate DRR into national and local development policies and planning.

3. Guiding Principles

The Sendai Framework is guided by several principles:

- **People-Centered:** Emphasizes the needs of people, particularly the most vulnerable, in disaster risk reduction efforts.
- **Inclusivity:** Ensures that all segments of society, including marginalized groups, are involved in DRR activities.
- **Sustainability:** Focuses on sustainable development and the integration of DRR into broader development agendas.
- **Risk Reduction as a Shared Responsibility:** Acknowledges that DRR is a shared responsibility of governments, private sector, and civil society.

4. Implementation and Monitoring

Implementation of the Sendai Framework involves:

- **National Action Plans:** Countries are encouraged to develop and implement national action plans aligned with the Framework.
- **Regional Cooperation:** Regional mechanisms and cooperation are vital for addressing transboundary risks and sharing best practices.
- **Monitoring Progress:** Regular monitoring and reporting mechanisms track progress toward achieving the global targets. The UN Office for Disaster Risk Reduction (UNDRR) plays a central role in monitoring and supporting implementation.

5. Linkages with Other Frameworks

The Sendai Framework complements and is linked to other global frameworks, including:

- **Sustainable Development Goals (SDGs):** Particularly Goal 11 (Sustainable Cities and Communities) and Goal 13 (Climate Action).
- **Paris Agreement on Climate Change:** Addresses the intersection of climate change and disaster risk reduction.
- **Agenda for Humanity:** Emphasizes humanitarian action and resilience-building.

6. Challenges and Opportunities

- **Challenges:** Effective implementation requires addressing gaps in data, ensuring adequate funding, and overcoming institutional barriers.
- **Opportunities:** The Framework presents opportunities for innovation in risk assessment, community engagement, and international collaboration.

The Stages of Disaster Risk Reduction

Disaster Risk Reduction (DRR) is a comprehensive process involving various stages aimed at reducing the impact of disasters. These stages encompass preparedness, mitigation, response,

and recovery. Each stage plays a crucial role in managing risks and ensuring a more resilient society. Here's a detailed overview of the stages of disaster risk reduction:

1. Risk Assessment and Analysis

Objective: Understand the hazards, vulnerabilities, and capacities to effectively plan and implement risk reduction measures.

Key Activities:

- **Hazard Identification:** Determine and characterize the potential hazards that could impact a community or region, such as natural disasters (earthquakes, floods, hurricanes) and human-induced events (industrial accidents).
- **Vulnerability Assessment:** Analyze the susceptibility of people, infrastructure, and systems to hazards. This includes evaluating factors like socioeconomic status, building structures, and access to resources.
- **Risk Analysis:** Combine hazard and vulnerability assessments to understand the potential impacts and likelihood of various disaster scenarios.

Outputs:

- Risk maps and profiles.
- Data on vulnerabilities and capacities.
- Forecasts and early warning systems.

2. Mitigation

Objective: Implement measures to reduce the severity or likelihood of disaster impacts before they occur.

Key Activities:

- **Structural Mitigation:** Enhance or modify infrastructure to withstand hazards, such as building earthquake-resistant structures or improving flood defenses.
- **Non-Structural Mitigation:** Develop and enforce regulations and policies, such as land-use planning, building codes, and environmental conservation practices.
- **Public Awareness and Education:** Promote understanding of risks and encourage community participation in DRR initiatives.

Outputs:

- Improved infrastructure and building standards.
- Policies and regulations supporting risk reduction.
- Community awareness campaigns and educational programs.

3. Preparedness

Objective: Ensure readiness to respond effectively to disasters when they occur.

Key Activities:

- **Emergency Planning:** Develop and regularly update emergency response plans, including evacuation routes, emergency shelters, and resource allocation.
- **Training and Exercises:** Conduct drills and training programs for emergency responders, community members, and other stakeholders to practice and refine response procedures.
- **Early Warning Systems:** Establish and maintain systems that provide timely and accurate alerts to communities about impending disasters.

Outputs:

- Comprehensive emergency response plans.
- Trained personnel and community members.
- Functional early warning systems.

4. Response

Objective: Provide immediate assistance to affected populations and manage the crisis during and immediately after a disaster.

Key Activities:

- **Emergency Response Coordination:** Mobilize and coordinate resources and personnel to address urgent needs, such as medical care, food, water, and shelter.
- **Communication:** Ensure effective communication among responders, affected communities, and the media to provide timely information and instructions.
- **Damage Assessment:** Conduct assessments to determine the extent of damage and prioritize response efforts.

Outputs:

- Timely and effective emergency assistance.
- Coordination among response agencies and organizations.
- Updated information on damage and needs.

5. Recovery

Objective: Restore and improve the affected community's infrastructure, services, and livelihood while addressing the underlying vulnerabilities that led to the disaster.

Key Activities:

- **Short-Term Recovery:** Focus on immediate needs such as restoring basic services, repairing damaged infrastructure, and providing financial and emotional support to affected individuals.
- **Long-Term Recovery:** Plan and implement strategies for rebuilding and improving resilience, including enhancing infrastructure, revising land-use policies, and strengthening community networks.
- **Evaluation and Learning:** Review the response and recovery efforts to identify lessons learned and best practices for future DRR initiatives.

Outputs:

- Restored and improved infrastructure and services.
- Enhanced community resilience and reduced vulnerabilities.
- Documented lessons learned and recommendations for future DRR efforts.

6. Review and Improvement

Objective: Continuously assess and enhance DRR strategies and practices based on experiences and evolving risks.

Key Activities:

- **Monitoring and Evaluation:** Regularly monitor the effectiveness of DRR measures and evaluate their impact on reducing disaster risks.
- **Updating Plans and Policies:** Revise and improve DRR plans, policies, and practices based on evaluation findings and changing risk profiles.
- **Community Feedback:** Engage with communities to gather feedback on DRR efforts and incorporate their insights into future strategies.

Outputs:

- Updated and refined DRR plans and policies.
- Improved DRR practices and strategies.
- Enhanced community engagement and feedback mechanisms.

Community-Based Disaster Risk Reduction (CBDRR)

Community-Based Disaster Risk Reduction (CBDRR) is an approach that involves engaging local communities directly in the process of identifying, assessing, and reducing disaster risks. Unlike top-down disaster management strategies, CBDRR emphasizes the active participation of the community members who are most vulnerable to disasters.

Definition and Principles

Community-Based: The approach is rooted in the belief that communities themselves are best positioned to understand their vulnerabilities and capacities. Community members are not just passive recipients of aid but active participants in reducing disaster risks.

Disaster Risk Reduction (DRR): This refers to strategies aimed at preventing or mitigating the impact of natural or human-induced disasters. DRR includes a range of activities, from building safer infrastructure to raising awareness and preparedness.

Core Components of CBDRR

Participatory Risk Assessment: Communities are involved in identifying and assessing the risks they face. This could include mapping out areas prone to flooding, understanding the impacts of climate change, or recognizing vulnerable groups within the community.

Capacity Building: Training and educating community members to develop their skills in disaster preparedness, response, and recovery. This includes first aid training, evacuation drills, and establishing community-based early warning systems.

Disaster Preparedness and Planning: Communities develop and implement disaster preparedness plans that are tailored to their specific needs. This might involve establishing community response teams, identifying evacuation routes, and creating resource inventories.

Mitigation and Prevention: Initiatives to reduce the potential impact of disasters. For example, planting mangroves to reduce coastal erosion, building flood-resistant infrastructure, or adopting sustainable agricultural practices to prevent soil degradation.

Advocacy and Policy Engagement: Communities engage with local governments and other stakeholders to advocate for policies and resources that support disaster risk reduction efforts.

Stages of CBDRR

Prevention and Mitigation: Focuses on reducing the risk of disaster through structural (e.g., building dikes) and non-structural (e.g., land-use planning) measures.

Preparedness: Ensuring that communities are ready to respond when a disaster strikes. This includes education, training, and developing contingency plans.

Response: Implementing the plans developed during the preparedness phase. Community members take the lead in responding to the disaster, supported by external agencies.

Recovery and Rehabilitation: After a disaster, communities are involved in rebuilding and restoring their lives, with a focus on reducing future risks.

Benefits of CBDRR

Empowerment: By involving communities in the decision-making process, CBDRR empowers them to take control of their own safety and resilience.

Local Knowledge Utilization: It leverages the indigenous knowledge and practices of communities, which are often well-suited to local conditions and risks.

Sustainability: CBDRR fosters long-term resilience as communities are more likely to maintain and adapt the measures, they have developed themselves.

Increased Social Cohesion: Working together on disaster risk reduction activities strengthens social ties and community solidarity.

Challenges

Resource Limitations: Communities, especially in developing regions, may lack the resources (financial, technical, etc.) to implement effective DRR measures.

Coordination with Authorities: Ensuring that community-based initiatives are aligned with broader governmental disaster management strategies can be challenging.

Sustainability: Without ongoing support and capacity building, community-led DRR efforts might falter over time.

Inclusivity: Ensuring that all segments of the community, including marginalized groups, are involved in the CBDRR process.

Examples

Village Disaster Management Committees (VDMCs): In many countries, communities have established VDMCs that coordinate local DRR activities, such as early warning systems and disaster drills.

Community-Based Flood Monitoring: In flood-prone areas, communities have developed local monitoring systems using simple tools to predict floods and disseminate warnings.

Resilient Agricultural Practices: Communities in drought-prone regions have adopted drought-resistant crops and water conservation techniques as part of their DRR strategy.

Community-Based Disaster Risk Reduction is a holistic approach that places communities at the heart of disaster management. By recognizing the unique risks and capacities of local populations, CBDRR fosters a culture of preparedness and resilience, ultimately leading to safer and more sustainable communities.

CBDRR IN COASTAL AREAS

Community-Based Disaster Risk Reduction (CBDRR) in coastal areas is an approach that emphasizes the active involvement of local communities in identifying, assessing, and reducing the risks associated with natural disasters. Coastal areas are particularly vulnerable to hazards such as cyclones, tsunamis, storm surges, coastal erosion, and rising sea levels due to climate change. CBDRR empowers communities to enhance their resilience and reduce the impact of such disasters. Here's a detailed explanation:

1. Community Engagement and Participation

Local Knowledge Integration: Coastal communities often possess valuable traditional knowledge about local hazards, weather patterns, and survival strategies. CBDRR involves integrating this knowledge with scientific data to create effective disaster risk reduction plans. This participatory approach ensures that the strategies developed are culturally appropriate and practical for the community.

Inclusive Planning: CBDRR ensures that all segments of the community, including women, children, the elderly, and marginalized groups, are involved in disaster risk reduction activities. Their perspectives and needs are considered in planning and implementation, leading to more comprehensive and equitable solutions.

2. Hazard and Vulnerability Assessment

Mapping and Risk Assessment: Communities are involved in mapping local hazards and identifying vulnerable areas. This includes assessing the risks of cyclones, flooding, storm surges, and erosion. Participatory mapping exercises help communities understand their vulnerabilities and prioritize areas that need attention.

Historical Analysis: By analyzing past disasters and their impacts, communities can identify patterns and develop strategies to mitigate future risks. This historical perspective, combined with current data, helps in creating a more informed and resilient disaster management plan.

3. Disaster Preparedness and Early Warning Systems

Community-Based Early Warning Systems: In coastal areas, timely warnings are crucial for saving lives. CBDRR promotes the establishment of community-based early warning systems that are managed and operated by local residents. These systems often involve traditional communication methods, such as flag signals or community radio, combined with modern technology like SMS alerts.

Evacuation Plans and Drills: Communities are trained to develop and regularly practice evacuation plans. Drills are conducted to ensure that everyone knows the safest routes and designated shelters in the event of a disaster. This preparedness reduces panic and confusion during actual emergencies.

4. Capacity Building and Training

Skills Development: CBDRR focuses on building the skills of community members to respond effectively to disasters. Training programs may include first aid, search and rescue, fire safety, and disaster management. These skills empower communities to take immediate action during emergencies, potentially reducing the impact of disasters.

Leadership Development: Community leaders are trained to coordinate disaster response efforts. Strong local leadership is essential for mobilizing resources, managing shelters, and ensuring effective communication during disasters.

5. Sustainable Livelihoods and Resource Management

Diversification of Livelihoods: In many coastal areas, communities rely heavily on fishing, agriculture, or tourism, which can be severely affected by disasters. CBDRR promotes the diversification of livelihoods to reduce dependence on a single income source. This may involve training in alternative livelihoods, such as eco-tourism, crafts, or aquaculture, which are less vulnerable to natural hazards.

Ecosystem-Based Approaches: Protecting and restoring natural ecosystems, such as mangroves, coral reefs, and sand dunes, is a key aspect of CBDRR in coastal areas. These natural barriers can reduce the impact of storm surges, erosion, and tsunamis. Communities are involved in activities like mangrove reforestation, which not only protects the coast but also provides additional resources like fish and timber.

6. Infrastructure Resilience and Risk Reduction

Building Resilient Structures: CBDRR involves the community in constructing disaster-resilient infrastructure. This includes elevated homes to avoid floodwaters, cyclone shelters, and sea walls. Local materials and techniques are often used, ensuring that structures are affordable and sustainable.

Retrofitting and Safe Building Practices: Existing buildings and infrastructure are assessed for vulnerability and retrofitted to withstand disasters. Communities are trained in safe building practices, which can significantly reduce the risk of damage during events like cyclones and floods.

7. Disaster Response and Recovery Planning

Community-Based Response Teams: Community members form disaster response teams that are activated during emergencies. These teams are responsible for first response activities, including rescuing trapped individuals, providing first aid, and managing shelters.

Post-Disaster Recovery and Reconstruction: CBDRR also focuses on long-term recovery and rebuilding. Communities are involved in planning and implementing reconstruction projects that are more resilient to future disasters. This includes rebuilding homes, restoring livelihoods, and rehabilitating ecosystems.

8. Policy Advocacy and Networking

Influencing Policy: Through CBDRR, communities can advocate for policies and practices that support disaster risk reduction. This may involve working with local governments to enforce building codes, secure funding for risk reduction projects, and integrate CBDRR into local development plans.

Networking and Partnerships: Communities are encouraged to build networks with other coastal communities, NGOs, and government agencies. These networks provide support, share knowledge, and strengthen the overall disaster management capacity of the region.

9. Education and Awareness

Disaster Risk Education: Regular awareness campaigns, workshops, and education programs are conducted to keep the community informed about the risks they face and how to mitigate them. Schools play a crucial role in educating children about disaster preparedness, ensuring that future generations are better equipped to handle emergencies.

Cultural Integration: Traditional practices and cultural knowledge are integrated into disaster risk education. This not only makes the information more relatable but also respects and preserves local heritage.

Community-Based Disaster Risk Reduction in coastal areas is a holistic approach that empowers communities to take charge of their own safety and resilience. By leveraging local knowledge, building capacity, and fostering collaboration, CBDRR enhances the ability of coastal communities to withstand and recover from disasters. This approach is crucial for sustainable development in vulnerable coastal regions, ensuring that communities are better prepared for the increasing risks posed by climate change and other hazards.

Disaster Cycle

Disaster management aims to reduce, or avoid the potential losses from hazards, assure prompt and appropriate assistance to victims of disaster, and achieve rapid and effective recovery. The disaster management cycle illustrates the ongoing process by which governments, businesses, and civil society plan for and reduce the impact of disasters, react during and immediately following a disaster, and take steps to recover after a disaster has occurred.

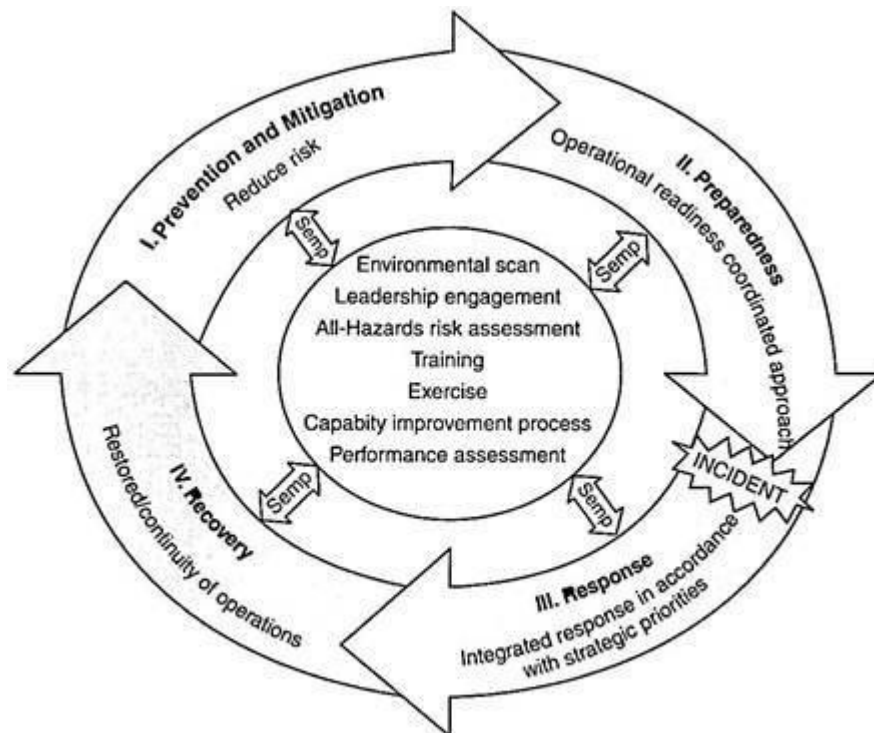


Fig. 2.7. The disaster management cycle

Appropriate actions at all points in the cycle lead to greater preparedness, better warnings, reduced vulnerability or the prevention of disasters during the next iteration of the cycle. The complete disaster management cycle includes the shaping of public policies and plans that either modify the causes of disasters or mitigate their effects on people, property, and infrastructure.

The mitigation and preparedness phases occur as disaster management improvements are made in anticipation of a disaster event. Developmental considerations play a key role in contributing to the mitigation and preparation of a community to effectively confront a disaster. As a disaster occurs, disaster management actors, in particular humanitarian organizations become involved in the immediate response and long-term recovery phases.

The four disaster management phases illustrated here do not always, or even generally, occur in isolation or in this precise order. Often phases of the cycle overlap and the length of each phase greatly depends on the severity of the disaster.

1. Mitigation:

Mitigation activities actually eliminate or reduce the probability of disaster occurrence, or reduce the effects of unavoidable disasters. Mitigation measures include building codes; vulnerability analyses updates; zoning and land use management; building use regulations and safety codes; preventive health care; and public education.

Mitigation will depend on the incorporation of appropriate measures in national and regional development planning. Its effectiveness will also depend on the availability of information on hazards, emergency risks, and the countermeasures to be taken. The mitigation phase, and indeed the whole disaster management cycle, includes the shaping of public policies and plans that either modify the causes of disasters or mitigate their effects on people, property, and infrastructure.

2. Preparedness:

The goal of emergency preparedness programs is to achieve a satisfactory level of readiness to respond to any emergency situation through programs that strengthen the technical and managerial capacity of governments, organizations, and communities. These measures can be described as logistical readiness to deal with disasters and can be enhanced by having response mechanisms and procedures, rehearsals, developing long-term and short-term strategies, public education and building early warning systems.

Preparedness can also take the form of ensuring that strategic reserves of food, equipment, water, medicines and other essentials are maintained in cases of national or local catastrophes. During the preparedness phase, governments, organizations, and individuals develop plans to save lives, minimize disaster damage, and enhance disaster response operations.

Preparedness measures include preparedness plans; emergency exercises/ training; warning systems; emergency communications systems; evacuations plans and training; resource inventories; emergency personnel/contact lists; mutual aid agreements; and public information/education.

As with mitigations efforts, preparedness actions depend on the incorporation of appropriate measures in national and regional development plans. In addition, their effectiveness depends on the availability of information on hazards, emergency risks and the countermeasures to be taken, and on the degree to which government agencies, non-governmental organizations and the general public are able to make use of this information.

During a disaster, humanitarian agencies are often called upon to deal with immediate response and recovery. To be able to respond effectively, these agencies must have experienced leaders, trained personnel, adequate transport and logistic support, appropriate communications, and guidelines for working in emergencies. If the necessary preparations have not been made, the humanitarian agencies will not be able to meet the immediate needs of the people.

3. Response:

The aim of emergency response is to provide immediate assistance to maintain life, improve health and support the morale of the affected population. Such assistance may range from providing specific but limited aid, range from providing specific but limited aid, such as assisting refugees with transport, temporary shelter, and food, to establishing semi-permanent settlement in camps and other locations.

It also may involve initial repairs to damaged infrastructure. The focus in the response phase is on meeting the basic needs of the people until more permanent and sustainable solutions can be found. Humanitarian organizations are often strongly present in this phase of the disaster management cycle.

4. Recovery:

As the emergency is brought under control, the affected population is capable of undertaking a growing number of activities aimed at restoring their lives and the infrastructure that supports them. There is no distinct point at which immediate relief changes into recovery and then into long-term sustainable development. There will be many opportunities during the recovery period to enhance prevention and increase preparedness, thus reducing vulnerability. Ideally, there should be a smooth transition from recovery to on-going development. Recovery activities continue until all systems return to normal or better. Recovery measures, both short and long term, include returning vital life-support systems to minimum operating standards; temporary housing; public information; health and safety education; reconstruction; counselling programs; and economic impact studies. Information resources and services include data collection related to rebuilding, and documentation of lessons learned.

Culture of Safety

Definition: A culture of safety involves creating an environment where safety is a fundamental value, and individuals are encouraged and empowered to take proactive steps to prevent and respond to hazards.

Components:

- **Leadership Commitment:** Leaders prioritize safety and demonstrate this through policies and actions.
- **Training and Education:** Regular training for employees and community members on safety protocols and procedures.
- **Open Communication:** Encouraging open dialogue about risks, safety concerns, and solutions.
- **Reporting and Feedback:** Implementing systems for reporting safety issues and incorporating feedback into safety practices.
- **Continuous Learning:** Staying updated on best practices and lessons learned from past incidents.

Benefits:

- **Enhanced Preparedness:** Individuals and organizations are more prepared for emergencies.
- **Reduced Risk:** Proactive measures reduce the likelihood and impact of accidents and disasters.
- **Improved Response:** A culture of safety supports effective and coordinated responses during emergencies.

Prevention, Mitigation, and Preparedness in Community-Based DRR

1. Prevention

- **Objective:** Avoid the occurrence of disasters where possible.
- **Activities:**
 - **Policy and Legislation:** Develop laws and regulations to prevent risky behaviours and practices.
 - **Community Engagement:** Involve communities in identifying and addressing potential risks.
 - **Education Programs:** Inform the public about actions they can take to prevent disasters.

2. Mitigation

- **Objective:** Reduce the severity and impact of disasters.
- **Activities:**
 - **Infrastructure Improvements:** Upgrade infrastructure to withstand hazards.
 - **Building Practices:** Implement construction methods that reduce vulnerability.
 - **Environmental Conservation:** Protect and restore natural systems that provide disaster protection (e.g., wetlands, forests).

3. Preparedness

- **Objective:** Ensure readiness for effective response and recovery.
- **Activities:**
 - **Community-Based DRR Plans:** Develop plans that involve community members in preparedness activities.

- o **Local Capacity Building:** Train local groups and organizations in disaster response and recovery.
- o **Early Warning Systems:** Establish and maintain systems to alert communities to imminent threats.

Structural and Non-structural Measures in Disaster Risk Reduction

In disaster risk reduction (DRR), strategies are often categorized into structural and non-structural measures. These two approaches complement each other in enhancing resilience against natural hazards. Here's a detailed exploration of both:

Structural Measures

Definition: Structural measures involve physical modifications or constructions aimed at reducing the impact of natural hazards. These measures directly alter the built environment to enhance its resilience against specific hazards.

Examples:

1. Seismic Design and Retrofitting:

- o **Seismic Isolation Systems:** These systems involve placing flexible bearings or isolators between a building's foundation and superstructure to absorb and dissipate earthquake energy, reducing the shaking felt by the building.
 - **Example:** The Taipei 101 skyscraper in Taiwan uses advanced seismic isolation techniques, including a large tuned mass damper, to protect the building from earthquakes and typhoons.
- o **Reinforced Concrete and Steel Frames:** Adding steel braces or reinforced concrete to existing structures enhances their ability to withstand seismic forces.
 - **Example:** After the 2011 Tohoku earthquake, many buildings in Japan underwent retrofitting with reinforced steel frames to improve their earthquake resilience.

2. Flood Control Infrastructure:

- o **Levees and Floodwalls:** These barriers prevent floodwaters from encroaching on urban areas and critical infrastructure.
 - **Example:** The Mississippi River Levees in the United States are a system of levees and floodwalls designed to protect communities along the river from flooding.
- o **Retention Ponds and Reservoirs:** These are designed to capture and store excess rainwater during heavy storms to prevent downstream flooding.
 - **Example:** The Central Park Reservoir in New York City helps manage stormwater runoff, reducing the risk of flooding in the surrounding areas.

3. Coastal Protection Structures:

- o **Seawalls:** Built along coastlines to protect against erosion and storm surges.
 - **Example:** The seawall in Miami Beach, Florida, protects the city from rising sea levels and storm surges, safeguarding properties and infrastructure.
- o **Breakwaters and Groynes:** Structures placed offshore or along coastlines to reduce wave energy and prevent coastal erosion.
 - **Example:** The breakwaters at the Port of Dunkirk in France protect the harbour from wave action and erosion.

4. Landslide Mitigation:

- o **Retaining Walls and Slope Stabilization:** Structures designed to prevent landslides by stabilizing slopes and retaining soil.
 - **Example:** In the Swiss Alps, retaining walls and mesh nets are used to stabilize slopes and prevent rockfalls that could affect roads and settlements.
 - o **Drainage Systems:** Installing systems to redirect water from slopes to prevent erosion and landslides.
 - **Example:** The installation of subsurface drainage systems in landslide-prone areas of Colombia helps manage water flow and reduce landslide risk.
5. **Fire-Resistant Buildings:**
- o **Fireproof Materials:** Using materials that resist fire can prevent buildings from burning during wildfires or urban fires.
 - **Example:** The use of fire-resistant materials like concrete and steel in the construction of buildings in wildfire-prone areas of California helps protect structures from wildfires.

Non-structural Measures

Definition: Non-structural measures involve strategies that do not involve physical construction but focus on policies, practices, and community engagement to reduce disaster risks and enhance preparedness.

Examples:

1. **Land-Use Planning and Zoning:**
 - o **Zoning Laws:** Regulating land use to prevent construction in high-risk areas and ensure safe development practices.
 - **Example:** In the floodplain areas of New Orleans, zoning regulations restrict new developments to minimize flood risk and ensure that existing structures are elevated.
 - o **Building Codes:** Establishing standards for construction to ensure buildings can withstand specific hazards.
 - **Example:** In Japan, building codes require earthquake-resistant designs, including flexible structures and reinforced materials, to enhance resilience against seismic activity.
2. **Public Education and Awareness:**
 - o **Disaster Preparedness Programs:** Educating the public about disaster risks and preparedness measures through workshops, drills, and information campaigns.
 - **Example:** The "Get Ready" campaign in New Zealand provides resources and workshops to help individuals and families prepare for earthquakes by creating emergency plans and kits.
 - o **School Drills and Training:** Conducting regular drills in schools to teach children how to respond during emergencies.
 - **Example:** In California, schools conduct earthquake drills to ensure students know how to "Drop, Cover, and Hold On" during an earthquake.
3. **Early Warning Systems:**
 - o **Tsunami Warning Systems:** Detecting seismic activity and issuing alerts to coastal communities to prepare for potential tsunamis.

- **Example:** The Pacific Tsunami Warning Center provides real-time tsunami alerts to countries around the Pacific Ocean, enabling timely evacuations.
- o **Weather Forecasting and Alerts:** Systems that provide timely information on severe weather conditions such as hurricanes, tornadoes, or floods.
 - **Example:** The National Weather Service in the United States issues weather warnings and watches via various channels, including radio, TV, and mobile alerts, to keep the public informed about imminent weather threats.
- 4. **Insurance and Financial Planning:**
 - o **Disaster Insurance:** Providing financial protection against property damage caused by disasters.
 - **Example:** The National Flood Insurance Program (NFIP) in the United States offers flood insurance to property owners in flood-prone areas, helping them recover financially from flood damage.
 - o **Emergency Fund Planning:** Encouraging individuals and businesses to set aside funds to cover immediate expenses following a disaster.
 - **Example:** Financial planning workshops often recommend creating emergency savings accounts to cover costs such as temporary housing and repairs after a disaster.
- 5. **Community-Based Disaster Risk Reduction (CBDRR):**
 - o **Local Risk Assessments:** Involving communities in identifying local hazards and vulnerabilities to develop targeted risk reduction strategies.
 - **Example:** In the Philippines, the Barangay Disaster Risk Reduction and Management Council (BDRRMC) engages local communities in identifying risks, creating evacuation plans, and conducting drills.
 - o **Participatory Mapping:** Engaging communities in creating maps that identify hazards, vulnerabilities, and resources to improve local disaster preparedness.
 - **Example:** In Indonesia, participatory mapping initiatives involve local communities in identifying flood-prone areas and creating maps that inform emergency planning and response.

Integration of Structural and Non-structural Measures

A comprehensive disaster risk reduction strategy integrates both structural and non-structural measures to create a robust framework for managing disaster risks. For instance, in a coastal city vulnerable to hurricanes, structural measures like seawalls and storm surge barriers can provide immediate physical protection, while non-structural measures such as zoning laws, public education, and early warning systems enhance overall preparedness and resilience.

Example of Integration: In a coastal city like New Orleans, a combination of structural and non-structural measures is used to manage flood risks. Structural measures include levees and floodwalls to physically block floodwaters, while non-structural measures involve zoning regulations to prevent construction in flood-prone areas, public awareness campaigns about emergency preparedness, and an integrated early warning system to alert residents about impending storms. This holistic approach ensures both immediate and long-term resilience against disasters.

By combining structural and non-structural measures, communities and organizations can effectively reduce disaster risks, enhance resilience, and improve overall preparedness and response capabilities.

Panchayati Raj Institutions

The Panchayati Raj Institutions (PRIs) form the backbone of Local Bodies in rural areas. These institutions aim to promote democratic participation, decentralization, and local self-governance. Let's take a closer look at the main components of PRIs:

Gram Panchayat

Gram Panchayat is the smallest unit of local self-government in rural areas. It consists of elected representatives known as Panchayat members, who serve as the interface between the local community and the government. The Gram Panchayat is responsible for various functions such as:

- o Providing essential services like water supply, sanitation, and healthcare at the village level.
- o Implementing government schemes and programs.
- o Promoting rural development and welfare measures.

Mandal Parishad/Panchayat Samiti

Mandal Parishad, also known as Panchayat Samiti, is the intermediate tier of Panchayati Raj Institutions. It comprises elected representatives from a group of Gram Panchayats within a block or mandal. The Mandal Parishad performs the following functions:

Coordination and supervision of the activities of Gram Panchayats within its jurisdiction.

- o Planning and implementing various development programs.
- o Allocating resources and finances to Gram Panchayats.

Zila Parishad/District Panchayat

Zila Parishad, or District Panchayat, is the apex body of the Panchayati Raj system at the district level. It consists of representatives elected from Mandal Parishads and members directly elected from constituencies within the district. The Zila Parishad carries out the following functions:

- o Overseeing and coordinating the activities of Mandal Parishads and Gram Panchayats.
- o Planning and implementing district-level development programs.
- o Managing funds and resources allocated to the district.

Roles and responsibilities of Panchayati Raj Institutions (PRIs)

Panchayati Raj Institutions (PRIs) play a fundamental role in disaster risk reduction (DRR) and management at the grassroots level in rural areas, functioning as the third tier of governance in many countries, such as India. **Their primary responsibility is to act as the local authority for implementing disaster management initiatives within their jurisdictions.** This includes preparing and executing disaster management plans that are tailored to the specific needs and risks of their communities. PRIs are tasked with identifying local hazards and vulnerabilities, conducting risk assessments, and developing contingency plans that include evacuation procedures, emergency response protocols, and community awareness programs.

During a disaster, PRIs are crucial in mobilizing local resources and coordinating immediate response efforts. They organize local response teams, set up temporary shelters, and ensure the distribution of emergency supplies such as food, water, and medical aid. PRIs work closely with state and central authorities to facilitate communication and ensure that the needs of affected populations are addressed promptly. Their deep knowledge of local conditions and networks enables them to effectively manage and direct local resources, enhancing the efficiency and effectiveness of emergency responses.

In the recovery phase, PRIs are pivotal in coordinating and overseeing reconstruction efforts within their areas. They facilitate the restoration of essential services such as water

supply, sanitation, and transportation, and help communities rebuild homes and infrastructure. PRIs also play a key role in managing and distributing recovery aid, including financial assistance and rebuilding materials. By engaging with local stakeholders and ensuring that recovery efforts align with community needs, PRIs contribute to the sustainable and equitable recovery of affected areas.

Furthermore, PRIs are instrumental in promoting disaster risk reduction and resilience building within their communities. They implement and support mitigation measures such as improving local infrastructure to withstand hazards, promoting sustainable land use practices, and organizing community-based training and awareness programs. PRIs encourage community participation in risk reduction activities, fostering a culture of preparedness and resilience. By integrating disaster risk reduction into local development plans and activities, PRIs help to reduce vulnerabilities and enhance the overall resilience of their communities.

Urban Local Bodies

Urban Local Bodies (ULBs) are counterparts of PRIs in urban areas. These bodies are responsible for governing cities and towns. Let's explore the different types of ULBs:

Nagar Panchayat

Nagar Panchayats are local bodies that govern small towns and areas in transition from rural to urban. They have limited resources and perform basic administrative functions such as:

- o Provision of water supply, street lighting, and sanitation services.
- o Maintenance of infrastructure like roads and parks.
- o Facilitating local-level development.

Municipal Council

Municipal Councils are local bodies that govern small and medium-sized cities. They have more extensive administrative powers compared to Nagar Panchayats. Some of their key functions include:

Providing essential services like water supply, electricity, waste management, and healthcare.
Ensuring urban planning and development.
Regulating trade, licenses, and permits.

Municipal Corporation

Municipal Corporations are the highest tier of Urban Local Bodies and are responsible for governing major cities and urban agglomerations. They have the most comprehensive administrative, financial, and executive powers. The functions of Municipal Corporations include:

- o Infrastructure development and maintenance.
- o Urban planning and land use management.
- o Administration of public services like education, healthcare, and transportation.

Role of Local Bodies

Local Bodies, whether in rural or urban areas, play a significant role in the governance and development of the country. Some of their essential roles are:

Decentralization: Local Bodies ensure decentralization of power, allowing governance to reach the local level.

Participation: They encourage citizen participation by involving the local population in decision-making processes.

Grassroots Development: Local Bodies address the specific needs and concerns of local communities, leading to targeted development initiatives.

Service Delivery: They are responsible for providing essential services like sanitation, healthcare, education, and infrastructure development.

Accountability: Local Bodies establish a direct link between the people and the government, promoting transparency and accountability.

Roles and responsibilities of Urban Local Bodies (ULBs)

Urban Local Bodies (ULBs) are crucial players in disaster risk reduction (DRR) and management, specifically within urban and semi-urban areas. As local governing institutions, ULBs encompass entities like municipal corporations, municipalities, and town councils, each responsible for managing urban areas and services. **Their primary role in disaster management involves preparing, implementing, and updating urban-specific disaster management plans.** These plans must address the unique risks and vulnerabilities associated with urban environments, such as infrastructure failures, high population density, and critical services like transportation and utilities.

In the preparedness phase, ULBs are tasked with developing and maintaining comprehensive emergency response strategies tailored to urban settings. This includes mapping out evacuation routes, setting up emergency shelters, and organizing community drills and awareness programs to ensure residents are well-informed and prepared. ULBs also collaborate with state and central authorities to integrate urban disaster management plans into broader regional and national frameworks, ensuring alignment and resource coordination.

During a disaster, ULBs play a pivotal role in managing and coordinating response efforts at the municipal level. They oversee the activation of emergency response protocols, deploy local emergency services, and coordinate the distribution of relief supplies such as food, water, and medical aid. ULBs are responsible for maintaining communication with higher levels of government to report on the situation, request additional resources if needed, and ensure that response operations are efficient and effective. Their proximity to the disaster-affected areas allows for rapid and localized action, crucial for managing urban emergencies.

In the recovery phase, ULBs are integral to overseeing the restoration of urban infrastructure and services. They manage the repair and rebuilding of critical facilities such as roads, bridges, water supply systems, and public buildings. ULBs also handle the distribution of recovery aid, including financial assistance for affected businesses and individuals, and coordinate with various stakeholders to ensure a comprehensive and equitable recovery process. By focusing on urban-specific recovery needs, ULBs help to restore normalcy and support the community's return to pre-disaster conditions.

Moreover, ULBs have a significant role in disaster risk reduction by implementing and enforcing urban planning and building regulations. They ensure that new constructions adhere to safety standards, promote resilient infrastructure, and integrate risk reduction measures into urban development plans. ULBs also engage in public education campaigns to raise awareness about disaster risks and encourage community participation in risk reduction

activities. By incorporating risk reduction strategies into urban planning and development, ULBs contribute to the long-term resilience of urban areas.

Roles and responsibilities of community

In disaster risk reduction (DRR) and management, the community plays a pivotal and multifaceted role, serving as the first line of defense and recovery in the face of disasters. At the preparedness stage, communities are instrumental in understanding local hazards and developing tailored emergency plans. These plans often include identifying evacuation routes, setting up emergency contact lists, and organizing community meetings to ensure that everyone is informed and prepared. Community members also participate in training sessions, such as first aid and basic firefighting, which equips them with the skills needed to respond effectively during emergencies.

During a disaster, the role of the community becomes even more critical. Community members provide immediate assistance to their neighbors, setting up local emergency shelters, distributing relief supplies, and participating in search and rescue operations. They are often the first responders on the ground, offering first aid and emotional support to those affected. Effective local response hinges on the community's ability to mobilize resources quickly and efficiently, leveraging local knowledge and networks to address immediate needs and stabilize the situation.

In the recovery phase, the community's involvement is crucial for rebuilding and restoring normalcy. Local residents contribute to the reconstruction of homes and infrastructure, repair damaged public facilities, and work to restore essential services. Community-led initiatives often drive recovery efforts, reflecting local priorities and ensuring that the recovery process aligns with the specific needs of the affected population. This local engagement helps to expedite recovery and rebuild social cohesion.

Moreover, communities are integral to disaster risk reduction by actively participating in risk assessments and implementing mitigation strategies. They engage in activities such as reinforcing homes, creating green spaces to manage flood risks, and adopting sustainable practices that reduce vulnerability to future hazards. By incorporating local knowledge into risk reduction planning and fostering a culture of preparedness, communities play a crucial role in enhancing their resilience to disasters.

Overall, the community's roles and responsibilities in disaster management encompass preparedness, immediate response, recovery, and risk reduction. Their active involvement ensures that disaster management efforts are grounded in local realities, effectively addressing immediate needs and contributing to long-term resilience. The strength and effectiveness of a community's response to disasters are directly linked to their level of preparedness and engagement, making their role indispensable in the broader framework of disaster risk reduction and management.

Roles and responsibilities of state

In disaster risk reduction (DRR) and management, state governments play a crucial and multifaceted role that bridges local and national efforts to ensure effective disaster preparedness, response, and recovery. **Firstly, at the preparedness stage, state governments are responsible for developing comprehensive disaster management plans that align with national frameworks while addressing specific regional hazards and vulnerabilities.** These plans outline procedures for emergency response, resource allocation, and coordination among various local and state agencies. The states also organize training

programs and simulation exercises for emergency responders, local officials, and community members to enhance their preparedness and response capabilities.

During a disaster, the state government's role involves significant coordination and resource management. States are tasked with overseeing and coordinating response efforts between local authorities and the central government, ensuring that resources are distributed effectively and that there is no duplication of efforts. They manage the deployment of additional resources, such as specialized response teams, equipment, and supplies, to support local and district-level operations. By serving as a central hub for communication and coordination, the state government ensures a streamlined and efficient response to emergencies.

In the recovery phase, the state government plays a pivotal role in facilitating and supporting long-term recovery efforts. This includes conducting damage and needs assessments to identify the extent of destruction and the required resources for recovery. Based on these assessments, states allocate financial aid and support for reconstruction efforts, which may involve grants, loans, and other forms of assistance for individuals, businesses, and local governments. They also oversee the implementation of recovery programs and initiatives, ensuring that they meet the needs of affected populations and promote sustainable rebuilding practices.

Additionally, state governments are responsible for implementing and enforcing policies related to disaster risk reduction. They develop and apply regulations and standards for building codes, land-use planning, and environmental management to minimize risks and enhance resilience. Investments in infrastructure projects, such as flood control systems and early warning systems, are also part of their responsibilities. By integrating risk reduction strategies into state policies and planning processes, states contribute to reducing vulnerabilities and improving overall disaster preparedness.

Roles and responsibilities of centre

In disaster risk reduction (DRR) and management, the central government holds a pivotal role, acting as the primary authority for setting national policies, coordinating large-scale responses, and facilitating long-term recovery efforts. **At the forefront of preparedness, the central government is responsible for establishing a comprehensive national disaster management framework that provides guidelines and standards for disaster risk reduction across the country.** This framework includes the development of national disaster management policies, regulations, and legislative measures that ensure a unified approach to disaster preparedness, response, and recovery. The central government also oversees the formulation of national plans that outline strategies for various types of hazards, ensuring that these plans are integrated with regional and local efforts.

During a disaster, the central government's role is crucial in coordinating and supporting national-level response efforts. It provides leadership and direction for managing large-scale emergencies that exceed the capacity of local and state resources. This includes mobilizing national resources, deploying specialized response teams such as the National Disaster Response Force (NDRF) in India, and coordinating with international organizations and agencies for additional support. The central government ensures that there is effective communication between national, state, and local levels, facilitating the flow of information and resources to where they are most needed.

In the recovery phase, the central government plays a key role in facilitating and managing financial and technical support for affected regions. It administers disaster relief funds and financial assistance programs, providing grants, loans, and other forms of aid to support reconstruction and rehabilitation efforts. The central government is also responsible for overseeing large-scale recovery initiatives, including infrastructure rebuilding, economic recovery programs, and social support services. By coordinating these efforts, the central government helps to ensure a cohesive and efficient recovery process that addresses the diverse needs of affected communities.

Furthermore, the central government is involved in promoting research and development in disaster risk reduction and management. It supports and funds research initiatives that advance understanding of disaster risks, improve mitigation technologies, and enhance early warning systems. The central government also fosters international cooperation, engaging with global frameworks and organizations to share knowledge, receive technical assistance, and contribute to international disaster risk management efforts.

Roles and responsibilities of stakeholders

In disaster risk reduction (DRR) and management, a wide array of stakeholders—each with distinct roles and responsibilities—contribute to a cohesive and effective approach to disaster preparedness, response, and recovery. **Communities are at the forefront, playing a crucial role in grassroots preparedness and response efforts.** They are responsible for developing localized emergency plans, conducting risk assessments, and organizing training and drills to ensure readiness for potential hazards. During emergencies, community members provide immediate assistance through local support networks, set up emergency shelters, and engage in search and rescue operations. Their active participation in recovery efforts, such as rebuilding infrastructure and restoring services, underscores their vital role in mitigating disaster impacts and enhancing resilience.

State governments act as the vital link between local communities and the central government, ensuring that disaster management is effective at a regional level. They are responsible for creating state-specific disaster management plans that integrate national policies with local needs. States coordinate the allocation of resources during emergencies, manage the deployment of additional support to affected areas, and oversee damage assessments to guide recovery efforts. Their role extends to implementing and enforcing regulations related to building codes, land use, and environmental management to reduce vulnerabilities and enhance resilience.

The central government provides overarching leadership and coordination for disaster management on a national scale. It establishes and enforces national disaster management policies, develops comprehensive frameworks for disaster response and recovery, and coordinates large-scale emergency responses. The central government is responsible for mobilizing national resources, deploying specialized response teams, and facilitating international assistance. In recovery, it administers disaster relief funds, supports infrastructure rebuilding, and promotes research and development in disaster risk reduction. Its role also includes fostering international cooperation and sharing knowledge to enhance global disaster management efforts.

Other stakeholders, including non-governmental organizations (NGOs), the private sector, and international bodies, play complementary roles in disaster management.

NGOs provide immediate relief services such as food, medical care, and shelter, and support long-term recovery by helping communities rebuild and restore livelihoods. The private sector contributes through business continuity planning, ensuring that companies can quickly resume operations after a disaster, and engages in Corporate Social Responsibility (CSR) initiatives that support disaster risk reduction projects. International organizations offer technical assistance, financial aid, and humanitarian support, enhancing national and local capacities to manage disasters. Academia and research institutions conduct vital research to improve understanding of disaster risks and develop effective mitigation strategies, while media outlets disseminate timely information and raise public awareness about disaster preparedness and response.

Early Warning System (EWS)

It can be defined as “An integrated system of hazard monitoring, forecasting and prediction, disaster risk assessment, communication and preparedness activities systems and processes that enable individuals, communities, governments, businesses and others to take timely action to reduce disaster risks in advance of hazardous events.”

- The early warning system is an adaptive measure for climate change, using integrated communication systems to help communities prepare for hazardous climate-related events.
- A successful EWS saves lives and jobs, land and infrastructures and supports long-term sustainability.
- Early warning systems will assist public officials and administrators in their planning, saving money in the long run and protecting economies.
- With human-induced climate change leading to more extreme weather conditions, the need for early warning systems is more crucial than ever.
- Systems that warn people of impending storms, floods or droughts are not a luxury but a cost-effective tool that saves lives, reduces economic losses, and provides a nearly tenfold return on investment.
- Early warning systems have helped decrease the number of deaths and have reduced losses and damages resulting from hazardous weather, water or climate events.
- But major gaps still exist, especially in small island developing states and least-developed countries.

Types of Early Warning Systems

Natural Disaster Warning Systems:

- **Meteorological Threats:** Systems that provide warnings for hurricanes, tornadoes, heavy rainfall, tsunamis, and other severe weather events, using satellite imagery, weather stations, radar, and other meteorological monitoring technologies.
- **Geological Threats:** Systems that alert populations about volcanic eruptions and earthquakes. These use seismographs, GPS, and other sensors to monitor earth movements and volcanic activity.
- **Hydrological Threats:** Systems designed to warn of flood risks from rising river levels or dam failures, utilizing water level gauges, flow sensors, and meteorological data.

Industrial and Technological Warning Systems:

- **Chemical Accidents:** Systems in industrial facilities that detect harmful chemical releases, typically involving sensor networks to monitor air quality and hazardous conditions.
- **Nuclear Incidents:** Systems that monitor radiation levels and operational parameters within nuclear facilities to detect and respond to anomalies that could lead to accidents.

Public Health Warning Systems:

- **Disease Outbreaks:** These systems use surveillance data to detect and respond to outbreaks of infectious diseases, enabling timely public health interventions.
- **Pandemic Alert Systems:** Global networks like the World Health Organization (WHO) monitor and provide alerts on emerging health threats that could reach pandemic proportions.

Components of an Effective Early Warning System

- **Detection and Monitoring:** Continuous monitoring using sensors and other data-collecting technologies to gather real-time information about potential hazards.
- **Data Analysis and Forecasting:** Analyzing the collected data to forecast potential impacts, using models and other predictive tools to assess when and where the disaster might strike.
- **Dissemination and Communication:** Efficient systems to communicate the warning to all stakeholders, including the government, emergency services, and the public. This may involve multiple channels like SMS, radio, TV, and internet-based communications to ensure broad coverage.
- **Response Capability:** Preparedness and emergency response plans that are well-practised and can be activated swiftly once a warning is issued. This includes training for emergency responders and public awareness campaigns.

Challenges

Despite the urgent need, only half of the countries worldwide report having adequate multi-hazard early warning systems.

And even fewer have regulatory frameworks that connect early warnings to emergency and response plans. There also are big gaps in the global observing system required to generate forecasts.

- **Accuracy and Timeliness:** Balancing the need for timely warnings with the risk of false alarms, can lead to complacency among the population.
- **Reach and Accessibility:** Ensuring that warnings reach all segments of the population, including vulnerable groups like the elderly, disabled, or those in remote areas.
- **Integration and Coordination:** Coordinating between different agencies and sectors to ensure a unified and effective response.

Climate, weather and water-related extremes have led to 15 times more deadly hazards for people in Africa, South Asia, South and Central America, and small island states.

- Vulnerable, least-developed countries that have not contributed significantly to the climate crisis are bearing the brunt.
- Over the last 50 years, nearly 70 per cent of all deaths from climate-related disasters have occurred in the 46 poorest countries.

Leveraging new technologies such as artificial intelligence (AI) and machine learning for better prediction and analysis, and using advanced communication tools to disseminate warnings more effectively should be prioritised.

Early Warning for All

Early Warnings for All is a groundbreaking initiative to ensure that everyone on Earth is protected from hazardous weather, water, or climate events through life-saving early warning systems by the end of 2027.

- As part of the UN Secretary-General's Acceleration Agenda, the Early Warnings for All initiative is a key contribution to delivering climate justice to those at the frontlines of the climate crisis.
- It aligns with the priorities of the Paris Agreement and supports key provisions of the Sendai Framework for Disaster Risk Reduction, particularly Target G on the availability and accessibility of multi-hazard early warning systems.
- It also contributes to delivering the targets of the 2030 Agenda for Sustainable Development on poverty, hunger, health, water, clean energy, climate action and sustainable cities.

Four pillars:

1. **Disaster risk knowledge and management:** Ensuring all countries have access to reliable, understandable and relevant risk information, science and expertise.
2. **Detection, observation, monitoring, analysis, and forecasting:** Ensuring all countries have robust forecast and monitoring systems (both soft and hardware infrastructure) and enabling policies to support optimization and sustainability of hazard monitoring and early warning systems.
3. **Warning dissemination and communication:** Using a people-centred approach to ensure that early warnings are effectively and timely disseminated to reach everyone, especially those most at risk.
4. **Preparedness and response capabilities:** Ensuring local governments, communities and individuals at risk know and means to take pre-emptive early actions to prepare for and respond to incoming disasters upon receiving warnings.

The Early Warnings for All initiative together the broader UN system, governments, civil society and development partners across the public and private sectors to enhance collaboration and accelerated action to address gaps and deliver people-centred, end-to-end multi-hazard early warning systems that leave no one behind.

Agencies involved in Disaster Management

- **National Disaster Management Authority (NDMA):** The National Disaster Management Authority, or the NDMA, is an apex body for disaster management, headed by the Prime Minister of India. It is responsible for the supervision, direction and control of the National Disaster Response Force (NDRF).
- **National Executive Committee (NEC):** The NEC is composed of high-profile ministerial members from the government of India that include the Union Home Secretary as Chairperson, and the Secretaries to the Government of India like Ministries/Departments of Agriculture, Atomic Energy, Defence, Drinking Water Supply, Environment and Forests etc. The NEC prepares the National Plan for Disaster Management as per the National Policy on Disaster Management.
- **State Disaster Management Authority (SDMA):** The Chief Minister of the respective state is the head of the SDMA. The State Government has a State Executive Committee (SEC) which assists the State Disaster Management Authority (SDMA) on Disaster Management.
- **District Disaster Management Authority (DDMA):** The DDMA is headed by the District Collector, Deputy Commissioner or District Magistrate depending on the situation, with the elected representatives of the local authority as the Co-Chairperson. The DDMA ensures that the guidelines framed by the NDMA and the SDMA are followed by all the departments of the State Government at the District level and the local authorities in the district.
- **Local Authorities:** Local authorities would include Panchayati Raj Institutions (PRI), Municipalities, District and Cantonment 11 Institutional and Legal Arrangements Boards, and Town Planning Authorities which control and manage civic services.

Advisories from appropriate agencies

Advisories from appropriate agencies are essential components of an effective Early Warning System (EWS) and disaster management framework. These advisories are formal communications issued by authoritative organizations that provide guidance, recommendations, or instructions to the public, government officials, and emergency responders regarding imminent or ongoing hazards. They are designed to inform and prompt specific actions to mitigate the impact of disasters, enhance preparedness, and coordinate response efforts.

The role of advisories is multifaceted and involves several key functions:

1. **Issuance of Alerts and Warnings:** Advisories are issued to alert communities about imminent threats or ongoing hazardous events. For instance, meteorological agencies might issue weather advisories for severe storms, heatwaves, or flooding, while geological agencies may issue earthquake or tsunami advisories. These advisories typically include information about the nature of the threat, its expected impact, and recommended actions to take. For example, a hurricane advisory might include details about the storm's path, expected wind speeds, and evacuation instructions.
2. **Providing Guidance and Recommendations:** Advisories offer actionable recommendations to help individuals and organizations prepare for and respond to disasters. This guidance can include practical measures such as securing property, stocking emergency supplies, or implementing safety protocols. For example, a flood advisory might recommend moving to higher ground, avoiding travel in affected areas, and securing valuable items.

3. **Facilitating Coordination and Response:** Advisories play a crucial role in coordinating response efforts among different stakeholders. They help synchronize actions between various agencies, local governments, and emergency services, ensuring a unified approach to disaster management. For instance, during a wildfire, a fire management advisory might coordinate with local authorities to implement evacuation plans, set up temporary shelters, and deploy firefighting resources.
4. **Enhancing Public Awareness and Preparedness:** By disseminating advisories, appropriate agencies educate the public about potential hazards and the steps they need to take to protect themselves. This includes raising awareness about risks and promoting preparedness practices. For instance, a public health advisory might provide information on how to prevent the spread of infectious diseases during a pandemic, including hygiene practices and vaccination guidelines.
5. **Supporting Recovery and Reconstruction:** After a disaster, advisories continue to play a role in the recovery process by providing information on available support services, rebuilding resources, and long-term safety measures. For example, a recovery advisory might include details about government aid programs, health services, and guidelines for rebuilding safely.

Different types of agencies involved in issuing advisories include:

- **Meteorological Agencies:** Such as the National Weather Service (NWS) or the Meteorological Department, which issue weather and climate-related advisories.
- **Geological Agencies:** Like the United States Geological Survey (USGS) or national seismic monitoring institutions, which issue advisories on earthquakes, tsunamis, and volcanic activity.
- **Health Departments:** Which provide advisories on public health issues, including disease outbreaks and epidemics.
- **Emergency Management Agencies:** Such as FEMA in the United States or local disaster management authorities, which coordinate response and recovery efforts.

Indigenous Knowledge

Indigenous knowledge refers to the traditional understanding, practices, and skills developed by local communities over generations. This knowledge is often deeply rooted in the cultural, environmental, and historical context of a community. In disaster management, indigenous knowledge provides valuable insights into local hazards, risks, and coping strategies. For example, indigenous communities may possess traditional methods for predicting weather patterns or recognizing signs of environmental changes that can indicate impending natural disasters like floods, storms, or earthquakes. They also often have established practices for managing and mitigating risks, such as constructing houses with materials and techniques suited to local conditions or employing traditional land management practices that prevent erosion and maintain soil fertility.

The relevance of indigenous knowledge lies in its deep connection to local conditions and its adaptability to specific environmental contexts. By integrating this knowledge with modern scientific approaches, disaster management strategies can be more culturally appropriate and effective. Indigenous practices often emphasize community involvement and local stewardship, which can enhance the effectiveness of disaster preparedness and response efforts. Moreover, respecting and incorporating indigenous knowledge helps

preserve cultural heritage and empower local communities, ensuring that their expertise is valued and utilized in disaster risk reduction initiatives.

Importance of Indigenous Knowledge.

Indigenous knowledge refers to the unique, traditional, and local understanding that has developed over centuries within specific communities, often passed down through generations. This knowledge is deeply rooted in the cultural practices, values, and experiences of indigenous peoples and is closely linked to their environment. When it comes to disaster management, indigenous knowledge plays a crucial role for several reasons:

1. Deep Understanding of Local Environment

Indigenous communities often have an intimate understanding of their local environment, including weather patterns, natural resources, and ecological changes. This knowledge allows them to predict and respond to natural events like floods, droughts, or cyclones, sometimes more accurately than modern forecasting methods.

For example, indigenous people in the Pacific islands have long relied on traditional weather signs, such as the behaviour of animals or the appearance of certain plants, to predict cyclones and prepare accordingly.

2. Traditional Practices in Disaster Risk Reduction

Indigenous knowledge includes a wealth of traditional practices that can mitigate disaster risks. These practices are often sustainable and adapted to the local environment, making them effective in reducing vulnerabilities.

Examples:

Terracing and Agroforestry: In mountainous regions, indigenous communities use terracing to prevent soil erosion and landslides during heavy rains. **Mangrove Planting:** Coastal communities have traditionally planted and maintained mangroves to protect against storm surges and coastal erosion. **Water Management Systems:** Indigenous methods of water harvesting and management, such as building small dams or using traditional irrigation systems, help in managing drought risks.

3. Community-Based Early Warning Systems

Indigenous knowledge often includes early warning systems that rely on natural indicators rather than technology. These systems are accessible to the entire community, including those who may not have access to modern communication tools. For instance, certain bird migrations or changes in animal behaviour have been traditionally used by indigenous groups to anticipate natural events like earthquakes or floods.

4. Sustainable Resource Management

Indigenous practices often emphasize the sustainable use of natural resources, which can reduce the impact of disasters. For example, traditional rotational farming and shifting cultivation methods help maintain soil fertility and prevent land degradation, thereby reducing the risk of drought and famine.

Sustainable fishing practices, as practiced by some indigenous coastal communities, ensure the long-term health of marine ecosystems, which can buffer against the impacts of tsunamis or storm surges.

5. Cultural Resilience and Social Cohesion

Indigenous knowledge fosters a strong sense of community and cultural identity, which can be critical in disaster response and recovery. The social networks and communal support systems inherent in indigenous cultures often facilitate quicker and more coordinated responses to disasters. Rituals, oral traditions, and communal gatherings also play a role in preserving and

disseminating disaster-related knowledge within the community, ensuring that younger generations are prepared for future risks.

6. Adaptation to Climate Change

As climate change intensifies the frequency and severity of natural disasters, indigenous knowledge offers valuable insights into adaptation strategies. Indigenous communities have long adapted to changing environmental conditions, and their knowledge can inform modern climate adaptation practices.

For example, the use of drought-resistant crop varieties, developed over generations by indigenous farmers, can help improve food security in the face of changing climate patterns.

7. Integration with Modern Science

Indigenous knowledge can complement modern scientific approaches to disaster management. By integrating traditional practices with contemporary technology, disaster management strategies can be more effective and culturally appropriate.

For instance, combining traditional floodplain management techniques with modern hydrological models can enhance flood risk reduction efforts.

8. Examples of Indigenous Knowledge in Disaster Management

The Maasai of East Africa: The Maasai people use their knowledge of livestock and the environment to manage droughts. They move their herds to different grazing areas based on traditional knowledge of water sources and pasture availability.

The Inuit of the Arctic: The Inuit use their knowledge of sea ice conditions, weather patterns, and animal behaviour to navigate safely in their harsh environment, which is increasingly important as climate change alters Arctic conditions.

9. Challenges in Utilizing Indigenous Knowledge

Documentation and Preservation: Much indigenous knowledge is oral and may be at risk of being lost as older generations pass away. Efforts are needed to document and preserve this knowledge.

Recognition and Respect: Indigenous knowledge is sometimes undervalued or overlooked by formal disaster management systems. Recognizing and integrating this knowledge requires respect for indigenous cultures and practices.

Collaboration with External Agencies: Effective use of indigenous knowledge in disaster management often requires collaboration between indigenous communities and external agencies, which can be challenging if there are differences in approach or communication.

10. Conclusion

Indigenous knowledge is an invaluable asset in disaster management, offering time-tested practices and insights that are deeply rooted in local environments. By integrating indigenous knowledge with modern disaster management strategies, communities can build more resilient systems that are both effective and culturally appropriate. Recognizing, preserving, and respecting this knowledge is essential for reducing.

Appropriate Technology

Appropriate technology refers to tools, techniques, and solutions that are specifically designed to be suitable for the local context, taking into account factors such as available resources, environmental conditions, and social needs. In disaster management, appropriate technology focuses on using simple, cost-effective, and sustainable solutions that can be easily adopted by local communities. For example, in flood-prone areas, the use of locally sourced materials to build elevated houses or barriers can help protect homes from water damage. Similarly, low-cost early warning systems using local communication methods can effectively alert communities to impending hazards.

The relevance of appropriate technology is its ability to address local needs without relying heavily on external resources or complex systems. It ensures that disaster management solutions are feasible, maintainable, and accessible for the communities that need them most. By focusing on technologies that are compatible with local conditions and resources, appropriate technology promotes sustainability and resilience, reduces dependency on external aid, and enhances the community's ability to manage and recover from disasters independently.

Local Resources

Local resources encompass the natural, human, and material assets available within a community. This includes natural resources like water, soil, and vegetation, as well as human resources such as local skills, knowledge, and labour. In disaster management, leveraging local resources can significantly enhance the effectiveness of preparedness and response efforts. For example, communities can use local materials to construct disaster-resistant infrastructure, employ local labour for emergency response and recovery tasks, and utilize natural resources for environmental management practices that reduce risk.

The relevance of local resources is their accessibility, affordability, and alignment with the community's needs and capacities. By relying on locally available resources, disaster management strategies can be more sustainable and cost-effective. Utilizing local resources also strengthens community ownership and involvement, as it encourages the use of familiar materials and practices. Furthermore, it reduces reliance on external supplies and support, enabling communities to respond more quickly and effectively in times of crisis.

Bhopal gas disaster (case study)

The Bhopal gas disaster, also known as the Bhopal tragedy, is one of the most devastating industrial accidents in history. Occurring on December 2-3, 1984, it involved the release of methyl isocyanate (MIC) gas from a pesticide plant owned by Union Carbide Corporation (UCC) in Bhopal, India. Here's a detailed case study of the disaster, including the causes, impacts, and lessons learned:

Background

Union Carbide India Limited (UCIL) operated a pesticide manufacturing plant in Bhopal, producing a chemical called Sevin, which contains methyl isocyanate. MIC is a highly toxic chemical used in the production of pesticides. The Bhopal plant was a key site for producing this compound.

Causes

1. Mechanical Failures:

- o The disaster was precipitated by the failure of a refrigeration unit designed to keep MIC at a low temperature to prevent its decomposition. The refrigeration system had been out of order for several weeks prior to the accident.

2. Operational Failures:

- o Safety systems, including gas scrubbers and vent gas systems, were not functioning effectively. Maintenance of critical safety equipment was inadequate, and some safety systems were disabled.

3. **Human Error:**
 - o Poor training of employees and inadequate response procedures contributed to the inability to manage the emergency effectively.
4. **Management Failures:**
 - o There was a lack of rigorous safety protocols and emergency response planning. The company's management had cut corners on safety and maintenance to reduce costs.
5. **Regulatory and Oversight Failures:**
 - o Inadequate regulatory oversight and enforcement by Indian authorities allowed unsafe practices to persist.

Immediate Impact

- **Release of Gas:** On the night of December 2-3, 1984, water entered a storage tank containing MIC, causing a violent chemical reaction. This reaction led to the release of a large cloud of MIC gas into the surrounding area.
- **Casualties:** The immediate release of the gas resulted in thousands of deaths within the first few hours. Estimates of the death toll vary, but it is generally accepted that around 3,000 to 4,000 people died in the immediate aftermath. Long-term health effects have caused additional deaths, with estimates suggesting up to 15,000 total deaths over the years.
- **Health Effects:** Survivors experienced severe respiratory problems, eye irritation, and long-term health issues including chronic respiratory diseases, cancer, and neurological disorders. The disaster also had significant psychological impacts on survivors.
- **Environmental Impact:** The surrounding environment was contaminated with toxic chemicals. Soil and water sources were polluted, leading to long-term ecological damage.

Long-term Consequences

- **Legal and Financial Repercussions:** Union Carbide faced significant legal challenges and financial liabilities. The company settled with the Indian government in 1989 for \$470 million, a sum widely criticized as insufficient to cover the damages and ongoing costs. Legal battles continued for years, with ongoing disputes over compensation and liability.
- **Health and Social Impact:** The health impacts on survivors have been profound, with continued medical and psychological support needed. Social structures in the affected area were disrupted, with many families losing their primary breadwinners and facing economic hardship.
- **Regulatory Changes:** The disaster led to significant changes in industrial safety regulations and disaster management practices in India. The Indian government introduced stricter industrial safety laws, and there was increased emphasis on emergency preparedness and response.

Lessons Learned

1. **Improved Safety Protocols:** The need for stringent safety protocols and regular maintenance in industrial operations became evident. Safety systems should be regularly tested and maintained to ensure they function correctly during emergencies.

2. **Emergency Preparedness:** Effective emergency response plans, including community evacuation plans and real-time communication systems, are critical. Training for both employees and local communities is essential.
3. **Regulatory Oversight:** Strong regulatory frameworks and enforcement are necessary to ensure compliance with safety standards. Regulatory bodies must be proactive in inspecting and overseeing industrial operations.
4. **Corporate Responsibility:** Companies must prioritize safety over cost-cutting measures. Ethical practices and corporate responsibility should be integral to business operations.
5. **Environmental and Public Health:** Long-term monitoring of environmental and public health impacts is crucial for managing the aftermath of industrial disasters. Adequate compensation and support for affected communities are essential for recovery.

The Bhopal gas disaster serves as a stark reminder of the potential consequences of industrial negligence and the importance of robust disaster management practices. Its legacy continues to influence industrial safety standards and emergency preparedness worldwide.

The Bhopal gas disaster of 1984 underscores the critical need for robust disaster management principles to prevent and mitigate such catastrophic events in the future.

Preparedness is fundamental; it involves rigorous training and awareness programs for both employees and the local community to ensure they are equipped to handle emergencies. Regular emergency drills, detailed response plans, and community engagement are crucial for effective preparedness.

Prevention focuses on enforcing stringent safety regulations, conducting thorough risk assessments, and implementing advanced safety systems in industrial facilities. This includes regular inspections, hazard analyses, and the use of fail-safes to minimize risks.

Mitigation involves reducing the potential impact through strategies like establishing buffer zones around hazardous sites, providing personal protective equipment, and maintaining public warning systems. These measures help limit the exposure and damage during an incident.

Response requires swift and coordinated action, including deploying emergency teams, ensuring effective evacuation procedures, and providing timely medical aid. Clear and accurate communication with the public is essential to manage the situation and provide necessary information.

Recovery efforts focus on both short-term and long-term support, including health care services, community rebuilding, and economic recovery. Learning from the disaster is crucial, involving post-disaster analysis to identify lessons learned, revising policies, and continuously improving safety practices and regulations. By integrating these principles of preparedness, prevention, mitigation, response, and recovery into disaster management frameworks, the impact of future industrial disasters can be significantly reduced, enhancing both safety and resilience.

